LABSHEET 2

1. Insert one record into the Student table.

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (1, 'Aman Sharma', 20, 'BCA', 'Meerut');
```

Explanation: Adds one student record.

2. Insert five student records using separate INSERT statements.

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (2, 'Riya Verma', 19, 'BCA', 'Delhi');

INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (3, 'Rahul Kumar', 21, 'BSc', 'Meerut');

INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (4, 'Neha Singh', 20, 'BCom', 'Roorkee');

INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (5, 'Arjun Malik', 22, 'BCA', 'Delhi');

INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (6, 'Priya Joshi', 19, 'BA', 'Dehradun');
```

Explanation: Each separate INSERT statement adds one row.

3. Insert multiple records into the Student table using a single statement.

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES
(7, 'Karan Mehta', 20, 'BCA', 'Meerut'),
(8, 'Simran Kaur', 21, 'BSc', 'Delhi'),
(9, 'Mohit Das', 19, 'BCom', 'Roorkee');
```

Explanation: One INSERT statement can add several rows.

4. Insert records into the Employee table.

```
INSERT INTO Employee (Emp_ID, Emp_Name, Department, Salary)
VALUES
(101, 'Vikas Sharma', 'IT', 45000.00),
(102, 'Pooja Verma', 'HR', 38000.00),
(103, 'Amit Singh', 'Sales', 42000.00);
```

Explanation: Adds three example employee records.

5. Insert records into the Department table.

```
INSERT INTO Department (Dept_ID, Dept_Name, Location)
VALUES
(10, 'IT', 'Delhi'),
(20, 'Human Resources', 'Roorkee'),
(30, 'Sales', 'Meerut');
```

Explanation: Adds three departments.

6. Insert records into the Product table.

```
INSERT INTO Product (Product_ID, Product_Name, Price, Quantity)
VALUES
(201, 'Keyboard', 799.00, 25),
(202, 'Mouse', 399.00, 40),
(203, 'Monitor', 8999.00, 10);
```

Explanation: Adds products with prices and stock quantities.

7. Insert records into the Customer table.

```
INSERT INTO Customer (Customer_ID, Customer_Name, Email, Phone)
VALUES
(301, 'Anjali Gupta', 'anjali@example.com', '9876543210'),
(302, 'Rohit Jain', 'rohit@example.com', '9876501234');
```

Explanation: Adds two example customer records.

8. Insert records into the Course table.

```
INSERT INTO Course (Course_ID,
Course_Name, Duration, Fees) VALUES
(401, 'BCA', '3 years', 60000.00),
(402, 'BSc Computer Science', '3
years', 45000.00), (403, 'MCA', '2
years', 80000.00);
```

Explanation: Adds three courses with their duration and fees.

9. Insert a student record with values for all columns.

```
INSERT INTO Student (Student_ID, Name,
Age, Course, City) VALUES (10, 'Kavita
Rawat', 20, 'BCA', 'Dehradun');
```

Explanation: Provides values for all five columns in the Student table.

10. Insert a student record by specifying only selected columns.

```
INSERT INTO Student (Student_ID, Name, Course)
VALUES (11, 'Dev Kumar', 'BCA');
```

Explanation: Omitted columns must allow NULL or have default values.

11. Update the city of a particular student.

```
UPDATE Student
SET City = 'Delhi'
WHERE Student_ID = 1;
```

Explanation: WHERE limits the update to student ID 1.

12. Update the salary of a particular employee.

```
UPDATE Employee
SET Salary = 50000.00
WHERE Emp_ID = 101;
```

Explanation: Changes the salary of employee 101.

13. Increase the salary of all employees by 10%.

```
UPDATE Employee
SET Salary = Salary * 1.10;
```

Explanation: Multiplies every salary by 1.10, increasing it by 10%.

14. Increase the price of all products by 5%.

```
UPDATE Product
SET Price = Price * 1.05;
```

Explanation: Increases each product price by 5%.

15. Change the course of a particular student.

```
UPDATE Student
SET Course = 'BSc'
WHERE Student_ID = 1;
```

Explanation: Changes the course for student ID 1.

16. Update the department of an employee.

```
UPDATE Employee
SET Department = 'Finance'
WHERE Emp_ID = 102;
```

Explanation: Changes employee 102's department to Finance.

17. Update the city of all students from Meerut to Delhi.

```
UPDATE Student
SET City = 'Delhi'
WHERE City = 'Meerut';
```

Explanation: Updates only students whose current city is Meerut.

18. Update the fees of a particular course.

```
UPDATE Course
SET Fees = 65000.00
WHERE Course_ID = 401;
```

Explanation: Changes the fees for course ID 401.

19. Update multiple columns of a student simultaneously.

```
UPDATE Student
SET Age = 21,
 Course = 'BCA',
 City = 'Roorkee'
WHERE Student_ID = 1;
```

Explanation: Changes three columns in one UPDATE statement.

20. Delete a particular student record using Student_ID.

```
DELETE FROM Student
WHERE Student_ID = 11;
```

Explanation: Deletes only the row with Student_ID 11.

21. Delete all students belonging to a particular city.

```
DELETE FROM Student
WHERE City = 'Roorkee';
```

Explanation: Deletes all rows where the city is Roorkee.

22. Delete employees whose salary is below a specified amount.

```
DELETE FROM Employee
WHERE Salary < 30000.00;
```

Explanation: Deletes employees whose salary is below 30,000.

23. Delete products whose quantity is zero.

```
DELETE FROM Product
WHERE Quantity = 0;
```

Explanation: Deletes products that have no stock.

24. Delete records satisfying multiple conditions using AND/OR.

```
-- AND: both conditions must be true
DELETE FROM Student
WHERE City = 'Delhi' AND Age < 20;

-- OR: either condition can be true
DELETE FROM Student
WHERE City = 'Meerut' OR City = 'Roorkee';
```

Explanation: Run one DELETE example at a time and check the condition first.

25. Complete INSERT, UPDATE and DELETE sequence on the Student table

Insert a sample student, display the table, update the record, display it again, delete the sample record, and display the table after deletion.

```
-- Step 1: Insert a sample student
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (999, 'Test Student', 20, 'BCA', 'Roorkee');

-- Step 2: Display the table after INSERT
SELECT * FROM Student;

-- Step 3: Update the student's city
UPDATE Student
SET City = 'Delhi'
WHERE Student_ID = 999;

-- Step 4: Display the table after UPDATE
SELECT * FROM Student;

-- Step 5: Delete the sample student
DELETE FROM Student




WHERE Student_ID = 999;

-- Step 6: Display the table after DELETE
SELECT * FROM Student;
```